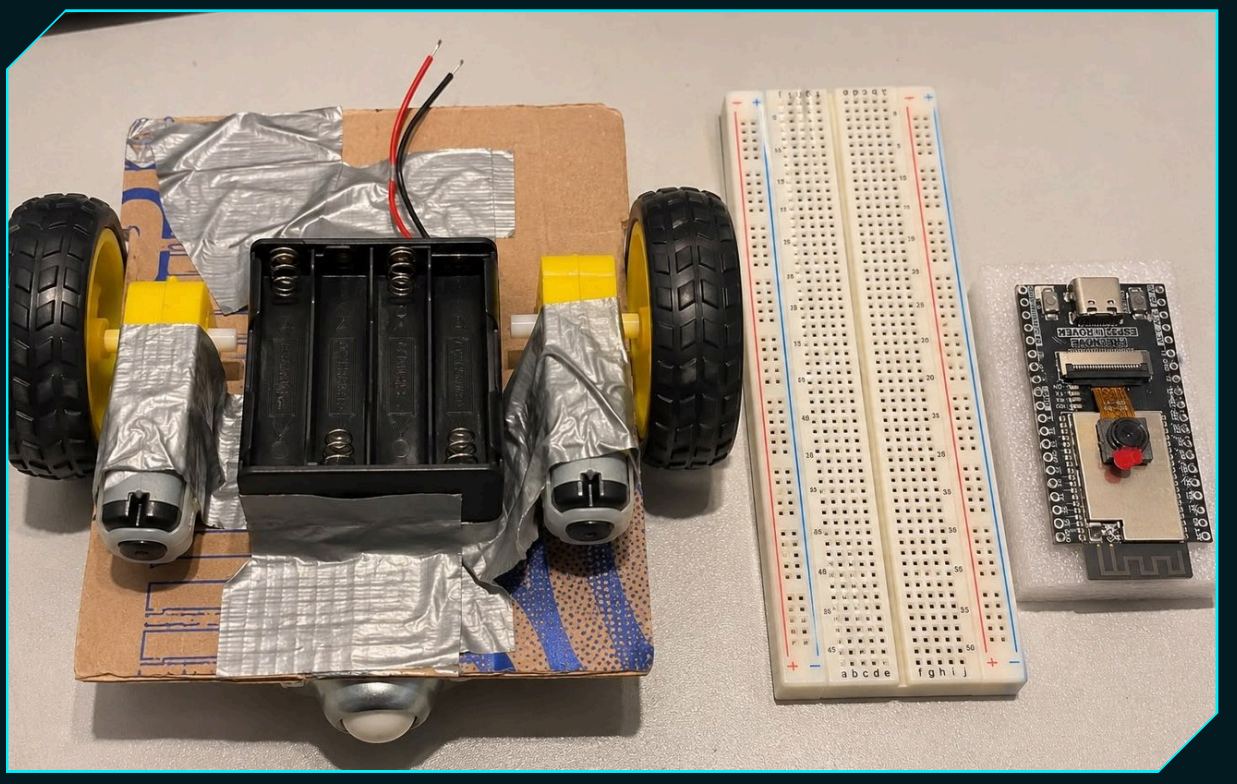
ROBOSOC

Obstacle-avoiding robot

WORKSHOP 5 · BUILD GUIDE



1
Building the chassis
steps 1–6

2
Electrical connections
steps 7–22

3
Final assembly
steps 23–30

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The code 16

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Wire types

`F-M`	female-to-male jumper
`F-M + joiner`	pin to pin: a female-to-male wire plus a joining wire we hand out (we have no female-to-female)
`M-M`	male-to-male jumper
🔴 🟠 🔵	3.3 V rail · 5 V rail · GND rail

Before you start

Parts

PART	QTY	STEPS
ESP32 development board	1	7–12
Yellow geared motors	2	2, 13
Wheels that fit the motors	2	3
Ball caster	1	4
Stiff cardboard (base, deck, strips)	—	1, 14, 27
Pmod DHB1 motor driver	1	10–13, 19
HC-SR04 ultrasonic sensor (5 V)	1	14–18
5 V regulator board	1	19–21, 29
4 × AA battery holder with switch	1	5, 19
AA batteries	4	20, 25
Breadboard (full size)	1	8, 16
330 Ω resistors	3	16
Jumper wires: female-to-male, a few male-to-male, plus joining wires we hand out	~20	8–18
Cable ties, electrical tape	—	2–5, 28
USB-A to USB-C data cable	1	23–26

Tools

- Small flat screwdriver, 2–3 mm
- Wire strippers and cutters
- Multimeter
- Soldering iron and solder
- Scissors, ruler, pencil
- Laptop with the Arduino IDE

Three rules

1. Batteries out, or switched off, while you wire.
2. Set the regulator to 5.0 V with a multimeter before it touches anything else.
3. The ESP32's 5 V pin gets the USB cable or the regulator, never both at once.

Left, right — As seen from behind the robot, front edge pointing away from you.

Rails — The long red + and blue – lines down each side of the breadboard. Step 8 sets them up.

How it works

About 30 times a second the robot measures the distance ahead, then decides what the wheels do. You can watch it happen in the Tinkercad simulation: [Obstacle-avoiding robot](#).

The parts and their jobs

PART	JOB
ESP32	Runs the code. Its pins work at 3.3 V.
HC-SR04	Sends a 40 kHz ping and times the echo.
Voltage divider	Drops the sensor's 5 V ECHO to 3.3 V for the ESP32.
Pmod DHB1	Switches battery power to the motors. An ESP32 pin can't power a motor.
5 V regulator	Turns 4 × AA into a steady 5 V for the ESP32 and sensor.
Motors	One per wheel. The robot steers by running them at different speeds.

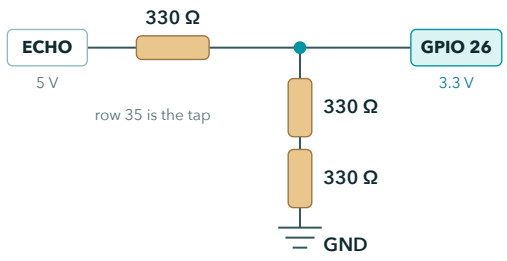
Measuring distance

```
cm = echo time in µs × 0.0343 ÷ 2
```

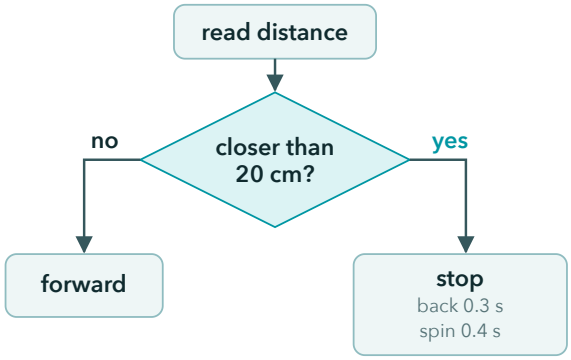
Sound travels 0.0343 cm every microsecond and goes there and back, so the time is halved. An echo of 1166 µs is 20 cm. No echo within 30 ms counts as 0, meaning nothing in range.

The voltage divider

One 330 Ω resistor above the tap and two below: 5 V × 660 ÷ 990 = 3.33 V. See it measured in Tinkercad: [Echo voltage divider](#).



Each loop



Driving a motor

DIR	EN	MOTOR
`0`	`PWM`	forward
`1`	`PWM`	reverse
`any`	`0`	stop

EN is switched on and off very fast (PWM). `analogWrite(EN1, 180)` keeps it on 70% of the time, so the motor runs at about 70% speed. **DIR** picks the direction. The code sets EN to 0 before changing DIR.

Turning

There is no steering. `drive(180, 180)` goes straight, `drive(180, -180)` spins right on the spot, `drive(180, 80)` curves right.

STAGE 1

1 Building the chassis

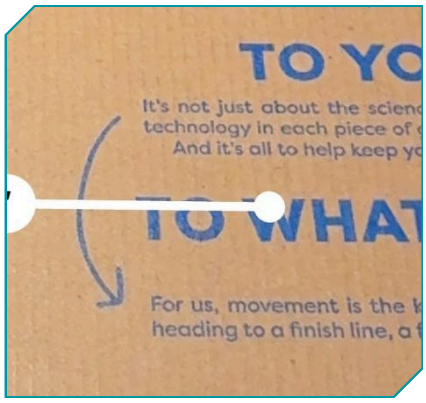
steps 1–6

FOR THIS STAGE Stiff cardboard · 2 yellow motors · 2 wheels · Ball caster · 4 × AA battery holder · Cable ties · Tape · Scissors

1 Prepare the base

YOU NEED Stiff cardboard · Ruler · Pencil · Scissors

- Cut a rectangle about 15 × 12 cm. It should not bend under the battery holder.
- Mark one short edge **FRONT**.



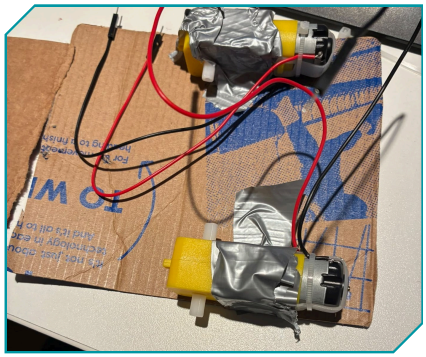
A box lid makes a good base.

2 Mount the motors

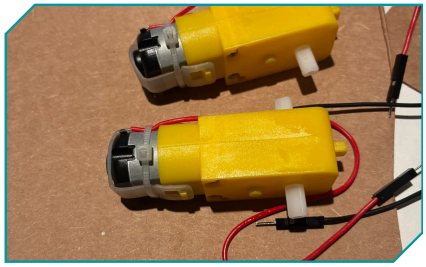
YOU NEED 2 yellow motors · 2 cable ties · Tape

- Lay the base flat with FRONT pointing away from you.
- One motor along the left edge, one along the right edge, about halfway along.
- Turn them so the white shafts point outwards, past the edges, in one straight line.
- Tape them in place.
- Poke two holes either side of each gearbox. Thread a cable tie down through one hole, under the board, up through the other and round the gearbox. Pull it tight.
- Keep the ties off the white shafts and the wires.

> **Check** Neither motor moves when you push it.



Right: opposite edges, shafts out.

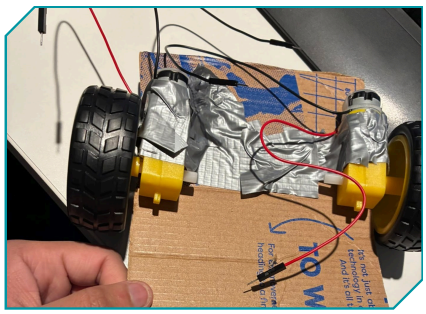


Wrong: both motors on one side.

3 Fit the wheels

YOU NEED 2 wheels

- Push a wheel straight onto each white shaft until it is snug.
- Spin each wheel by hand. Nothing should rub on the cardboard or the ties.



4 Fit the ball caster

YOU NEED Ball caster · 1 cable tie

- Turn the base over. Place the caster at the front, in the centre, ball facing out.
- Mark its two mounting holes and poke matching holes through the cardboard.
- Thread a cable tie up through one cardboard hole and the caster's first hole, across the metal plate (not over the ball), down through its second hole and back through the cardboard. Lock it and trim it.
- Turn the robot upright.

> **Check** Both wheels and the caster touch the table. If the caster hangs in the air, pack cardboard under its plate.



Threading the tie.

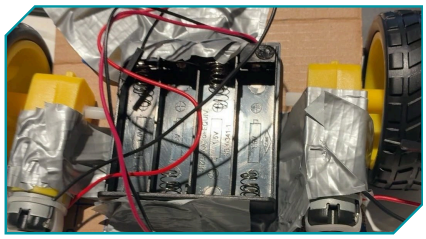


Done: tie across the plate.

5 Fit the battery holder

YOU NEED 4 × AA battery holder · Tape

- Tape the holder on top, in the middle, between the motors.
- Leave its red and black wires free and the battery slots reachable.
- No batteries yet.
- Keep bare wire ends away from metal foil tape.

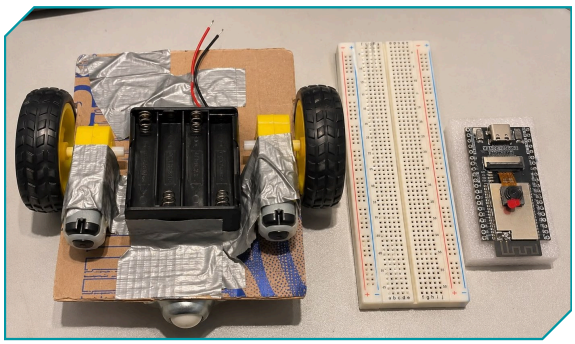




6 Lay out the electronics

YOU NEED Breadboard · ESP32 · Motor driver

- The breadboard is longer than the chassis. Do all the wiring on the desk, beside the robot.
- Everything moves onto an upper deck in step 27, once it is tested.



Finished chassis, electronics beside it.

2 STAGE 2 Electrical connections

steps 7–22

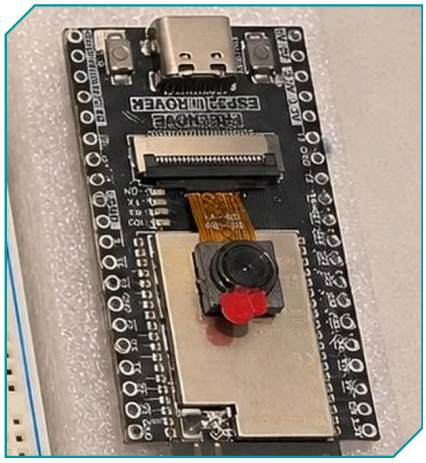
FOR THIS STAGE ESP32 · Pmod DHB1 motor driver · Breadboard · HC-SR04 sensor · 3 × 330 Ω resistors · 5 V regulator · Jumper wires · Small screwdriver · Wire strippers · Multimeter · Soldering iron

ESP32 and motor driver steps 7–13

7 Check the ESP32

YOU NEED ESP32

- Look underneath: both long edges need metal header pins. Solder them on if the holes are empty.
- **Don't put it in the breadboard.** It lies next to it, and every connection is a jumper wire pushed onto one of its pins.
- Pins are labelled by number. `13` means GPIO 13.



Beside the breadboard, not in it. Header pins on both long edges.

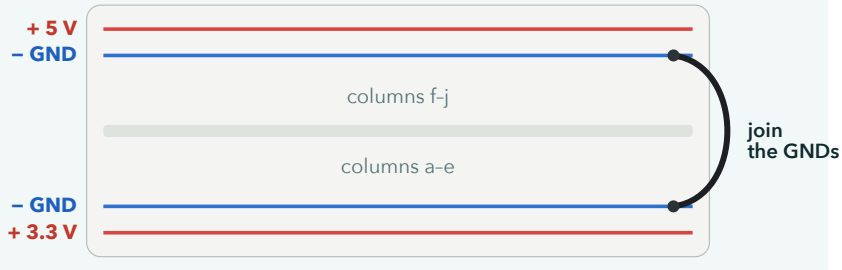
8 Set up the breadboard rails

YOU NEED Breadboard · 1 male-to-male jumper

RAIL	WHERE	FOR
3.3 V rail	red + line beside column a	`driver logic`
5 V rail	red + line beside column j	`ESP32 5V pin, sensor`
GND rail	both blue − lines	`every ground`

FROM	TO	WIRE
blue − line, column a side	blue − line, column j side	`M-M`

- If a rail line has a gap in the middle, bridge it or keep everything on one half.

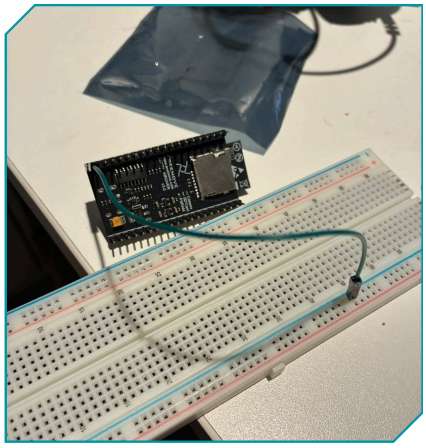


9 Power the rails from the ESP32

YOU NEED 3 female-to-male jumpers

FROM	TO	WIRE
ESP32 **GND**	GND rail	F–M
ESP32 **3V3**	3.3 V rail	F–M
ESP32 **5V**	5 V rail	F–M

Keep the two red rails apart. 5 V on the 3.3 V rail can damage the boards.



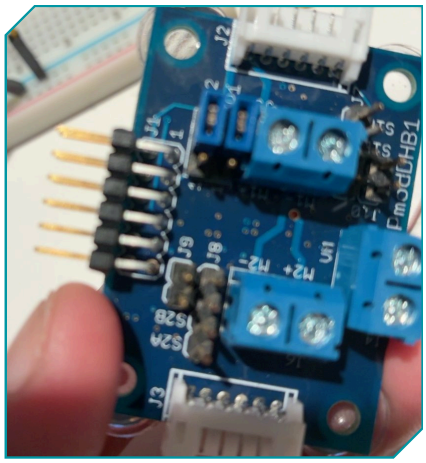
First wire: ESP32 GND to the rail.

10 Find J1 and set the jumpers

YOU NEED Motor driver

- J1 is the 12 gold pins in two rows of six. Find the small **1** printed beside it: that is pin 1. Pin 7 sits directly below it.
- Check jumpers **JP1** and **JP2** are both on position 3. On position 1 the right motor copies the left one.

1	2	3	4	5	6
EN1 GPIO 13	DIR1 GPIO 14	S1A	S1B	GND GND rail	VCC 3.3 V rail
pin 7 sits directly under pin 1					
EN2 GPIO 32	DIR2 GPIO 33	S2A	S2B	GND	VCC
7	8	9	10	11	12



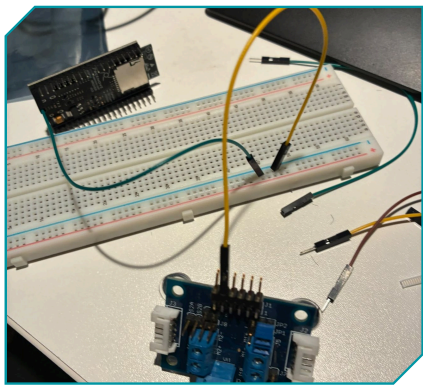
J1 on the left edge.

11 Driver power

YOU NEED 2 female-to-male jumpers

FROM	TO	WIRE
J1 pin 5 **GND**	GND rail	F–M
J1 pin 6 **VCC**	3.3 V rail	F–M

Check Pin 6 is next to pin 5. Make sure GND and VCC are not swapped.



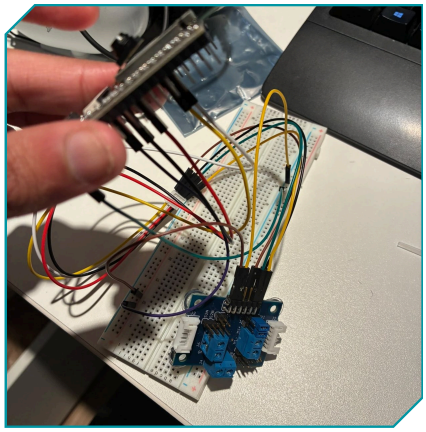
12 Control wires

YOU NEED 4 female-to-male jumpers · 4 joining wires

FROM	TO	WIRE
J1 pin 1 **EN1**	ESP32 **13**	`F-M + joiner`
J1 pin 2 **DIR1**	ESP32 **14**	`F-M + joiner`
J1 pin 7 **EN2**	ESP32 **32**	`F-M + joiner`
J1 pin 8 **DIR2**	ESP32 **33**	`F-M + joiner`

- Pin to pin, with no female-to-female wires: push a female-to-male wire onto the J1 pin, then connect its male end to the ESP32 pin with one of the joining wires we hand out.

Check 14, not 4.



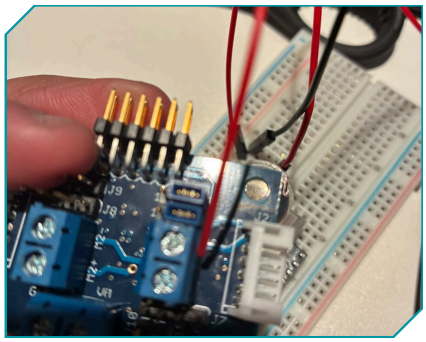
13 Connect the motors

YOU NEED Small flat screwdriver · Wire strippers

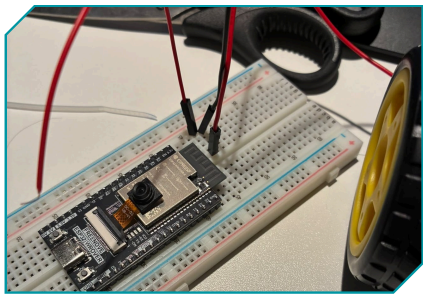
- Cut off the pin ends of the motor leads and strip about 5 mm of insulation.
- Turn a terminal screw 2–3 turns anticlockwise, push the bare copper into the square hole on the side of the blue block, then tighten clockwise. Tug to check.

FROM	TO	WIRE
Left motor **red**	J5 **M1+**	`motor lead`
Left motor **black**	J5 **M1−**	`motor lead`
Right motor **red**	J6 **M2+**	`motor lead`
Right motor **black**	J6 **M2−**	`motor lead`

Check No loose strands touch the next terminal. A wheel that turns the wrong way gets fixed in code later.



Leads clamped in the terminal.



The leads come with pin ends: cut them off.

Ultrasonic sensor `steps 14–18`

14 Mount the sensor

YOU NEED HC-SR04 · Cardboard strip 6 × 4 cm · Tape

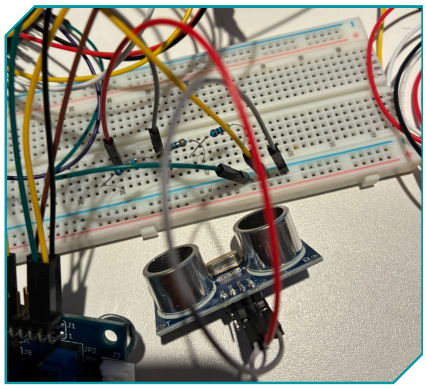
- Fold the cardboard strip into an L.
- Tape the flat part on top of the base at the front edge, in the centre.
- Tape the HC-SR04 to the upright part, silver cylinders facing straight ahead and level.
- Keep tape off the cylinders and leave the four pins reachable.

15 Sensor ground and trigger

YOU NEED 2 female-to-male jumpers · 1 joining wire

FROM	TO	WIRE
HC-SR04 **GND**	● GND rail	`F–M`
HC-SR04 **TRIG**	ESP32 **25**	`F–M + joiner`

- TRIG to the ESP32 is pin to pin: same as step 12, a female-to-male wire plus a joining wire.



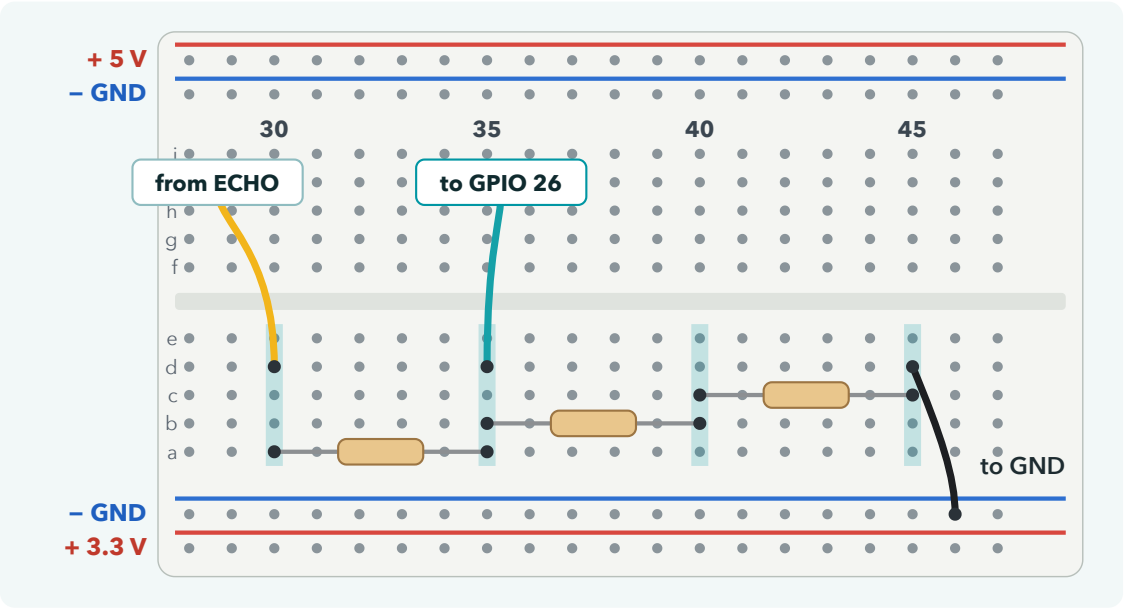
16 Build the echo divider

YOU NEED 3 × 330 Ω resistors

	FROM	TO
Resistor 1	a30	`a35`
Resistor 2	b35	`b40`
Resistor 3	c40	`c45`

- Want to see it first? The same divider in Tinkercad: [Echo voltage divider].

> **Check** Each resistor bridges two different rows. Never both legs in one row.



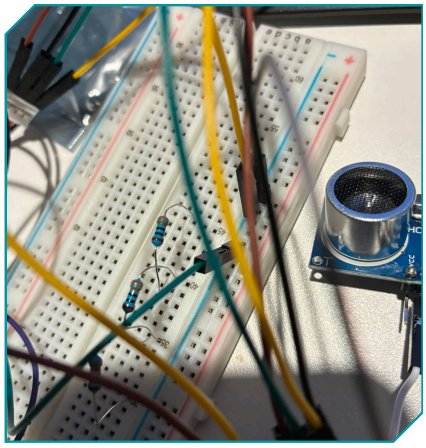
Rows 30–45. Yellow: from ECHO. Teal: to GPIO 26. Black: to GND.

17 Wire the divider

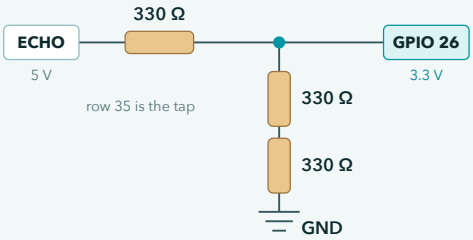
YOU NEED 2 female-to-male jumpers · 1 male-to-male jumper

FROM	TO	WIRE
HC-SR04 **ECHO**	breadboard **d30**	F–M
breadboard **d35**	ESP32 **26**	F–M
breadboard **d45**	● GND rail	M–M

ECHO never goes straight to the ESP32. It puts out 5 V; the ESP32 takes 3.3 V. The divider brings it down to 3.3 V.



The three resistors in a chain.



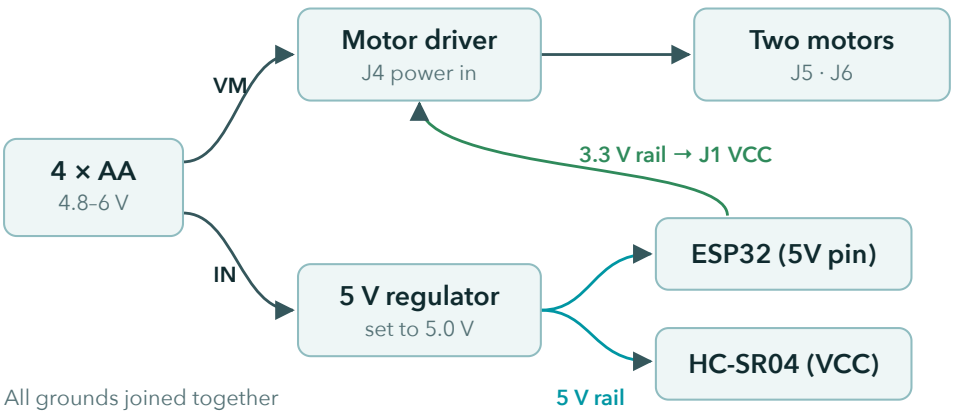
18 Sensor power

YOU NEED 1 female-to-male jumper

FROM	TO	WIRE
HC-SR04 **VCC**	● 5 V rail	F–M

Check VCC is on the 5 V rail, not the 3.3 V rail.

Power steps 19–22

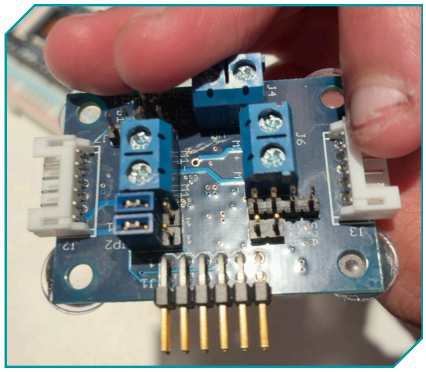


19 Batteries and regulator into J4

YOU NEED Battery holder · 5 V regulator · Soldering iron · Screwdriver

- Batteries out.
- If the regulator has bare pads, solder a short wire to each of IN+, IN−, OUT+ and OUT−. The arrow on the back points from IN to OUT.
- Twist each battery wire together with the matching regulator IN wire, then clamp the pair in J4:

FROM	TO	WIRE
Battery **red** + regulator **IN+**	J4 **VM**	`twisted pair`
Battery **black** + regulator **IN−**	J4 **GND**	`twisted pair`

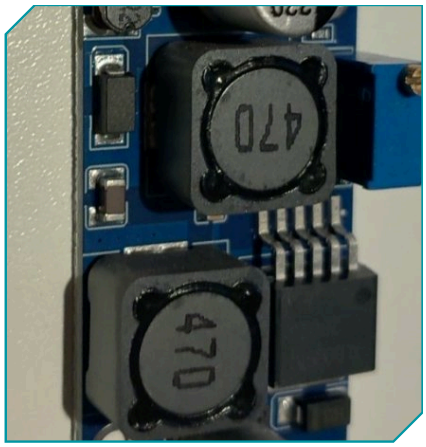


J4 is the blue terminal at the top.

20 Set the regulator to 5.0 V

YOU NEED 4 AA batteries · Multimeter · Small screwdriver

- OUT+ and OUT− stay connected to nothing for now.
- Put the batteries in and switch on.
- Multimeter on DC volts: red probe on OUT+, black on OUT−.
- Turn the brass screw on the blue trimmer until it reads 5.0 V. It can take several turns before the reading moves.
- Switch off.



The brass screw sets the output.

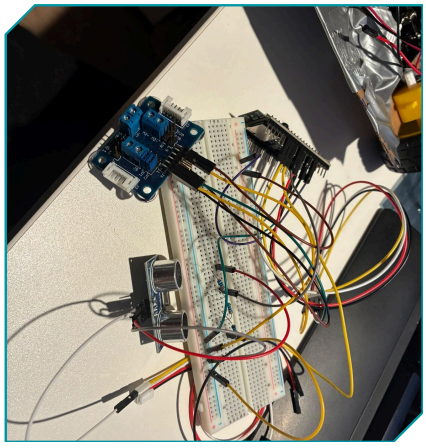
21 Regulator ground

FROM	TO	WIRE
Regulator **OUT−**	● GND rail	`soldered lead`

- Tape over the bare end of OUT+ and leave it loose. It goes in at step 29.

22 Check everything

- Batteries off, USB unplugged.
- Go through **Wiring at a glance** at the back of this guide, one line at a time.
- Look for bare copper touching anything it shouldn't.



Everything wired on the desk.

3 STAGE 3 Final assembly

steps 23–30

FOR THIS STAGE Laptop with Arduino IDE · USB-C data cable · A box to stand the robot on · Cardboard 18 × 12 cm · Cardboard strips · Tape · Cable ties

Code and tests steps 23–26

23 Set up the Arduino IDE

YOU NEED Laptop · USB-C data cable

- Install the Arduino IDE from arduino.cc.
- Boards Manager: install **esp32 by Espressif Systems**.
- Open `obstacle_robot.ino` (full listing at the back).
- Tools → Board → esp32 → **ESP32 Wrover Module**. Tools → Port → the port that appears when you plug the board in.
- No port at all? Use a data cable, and install the CH340 USB driver.

24 Test 1: sensor

- Batteries off. USB in.
- Set `const int MODE = 1;` and upload.
- Open the Serial Monitor at 115200 baud.
- Hold your hand 10, 20 and 30 cm in front of the sensor.

> **Check** The readings match the real distance within a couple of cm.

25 Test 2: motors

YOU NEED A box

- Stand the robot on a box so the wheels spin in the air.
- Set `MODE = 2`, upload, then switch the batteries on.
- Watch: left forward, right forward, both backward, stop.
- A wheel going the wrong way: set `FLIP_LEFT` or `FLIP_RIGHT` to `true` and upload again.
- Switch the batteries off.

26 Test 3: drive code

- Still on the box: set `MODE = 3`, upload, switch the batteries on.
- Both wheels turn forward. Put a hand in front of the sensor: they stop, reverse, then spin.
- Switch off and unplug USB.
- Robot misbehaving? Compare with the Tinkercad simulation of the same logic: **Obstacle-avoiding robot**.

Mount and drive `steps 27–30`

27 Build the upper deck

YOU NEED Cardboard 18 × 12 cm · Cardboard strips · Tape

- Make four pillars about 4 cm tall: fold cardboard strips into square tubes and tape them.
- Tape a pillar at each corner around the battery holder, then tape the deck on top.
- Keep the deck behind the sensor and leave the battery switch reachable.

28 Move the electronics up

YOU NEED Tape · Cable ties

- Tape the breadboard and the motor driver onto the deck.
- Rest the ESP32 on two cardboard strips so its pins and wires hang free, then tape the strips down.
- Tie loose wires away from the wheels. Refit any wire that came out.

29 Go battery-powered

- USB unplugged.

FROM	TO	WIRE
Regulator **OUT+**	5 V rail	`soldered lead`

- Put the robot on the floor and switch on. It waits 2 seconds, then drives.

30 Tune it

- Change `SPEED`, `STOP_DISTANCE`, `BACK_TIME` and `TURN_TIME` at the top of the code.
- To upload again: switch off, pull OUT+ off the 5 V rail, then plug in USB. Undo both before driving.

REFERENCE

The code

obstacle_robot.ino

Change `MODE` for each test in steps 24–26. The settings under "Tune these" are the ones to play with.

```
// RoboSoc Workshop 5: obstacle-avoiding robot
// ESP32 + Pmod DHB1 motor driver + HC-SR04 ultrasonic sensor

// 1 = sensor test, 2 = motor test, 3 = drive
const int MODE = 1;

// Pmod DHB1, header J1
const int EN1  = 13;  // J1 pin 1: left motor on/speed
const int DIR1 = 14;  // J1 pin 2: left motor direction
const int EN2  = 32;  // J1 pin 7: right motor on/speed
const int DIR2 = 33;  // J1 pin 8: right motor direction

// HC-SR04
const int TRIG = 25;
const int ECHO = 26;  // through the resistor divider

// Tune these
const int SPEED         = 180;  // 0 to 255
const int STOP_DISTANCE = 20;   // cm
const int BACK_TIME     = 300;  // ms
const int TURN_TIME     = 400;  // ms

// Wheel going the wrong way in the motor test? Change its false to true.
const bool FLIP_LEFT  = false;
const bool FLIP_RIGHT = false;

void setup() {
  Serial.begin(115200);
  pinMode(EN1, OUTPUT);
  pinMode(DIR1, OUTPUT);
  pinMode(EN2, OUTPUT);
  pinMode(DIR2, OUTPUT);
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  stopMotors();
  delay(2000);
}

void loop() {
  if (MODE == 1) sensorTest();
  if (MODE == 2) motorTest();
  if (MODE == 3) avoidObstacles();
}

void avoidObstacles() {
  int distance = readDistance();

  if (distance > 0 && distance < STOP_DISTANCE) {
    stopMotors();
    delay(100);
    drive(-SPEED, -SPEED);  // back up
    delay(BACK_TIME);
    drive(SPEED, -SPEED);   // spin right
```

```
    delay(TURN_TIME);
  } else {
    drive(SPEED, SPEED);  // forward
  }
  delay(20);
}

void sensorTest() {
  Serial.print("Distance: ");
  Serial.print(readDistance());
  Serial.println(" cm");
  delay(200);
}

void motorTest() {
  Serial.println("Left wheel forward");
  drive(SPEED, 0);
  delay(1500);
  Serial.println("Right wheel forward");
  drive(0, SPEED);
  delay(1500);
  Serial.println("Both wheels backward");
  drive(-SPEED, -SPEED);
  delay(1500);
  Serial.println("Stop");
  stopMotors();
  delay(3000);
}

// Distance in cm. 0 means no echo came back.
int readDistance() {
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);

  long echoTime = pulseIn(ECHO, HIGH, 30000);  // microseconds
  return round(echoTime * 0.0343 / 2);
}

// left and right: -255 (full reverse) to 255 (full forward)
void drive(int left, int right) {
  if (FLIP_LEFT)  left  = -left;
  if (FLIP_RIGHT) right = -right;
  setMotor(EN1, DIR1, left);
  setMotor(EN2, DIR2, right);
}

void setMotor(int enPin, int dirPin, int speed) {
  analogWrite(enPin, 0);  // stop before changing direction
  digitalWrite(dirPin, speed < 0 ? HIGH : LOW);
  analogWrite(enPin, abs(speed));
}

void stopMotors() {
  analogWrite(EN1, 0);
  analogWrite(EN2, 0);
}
```

REFERENCE

Wiring at a glance

FROM	TO	STEP
Rails and ESP32		
Blue – line (column a side)	blue – line (column j side)	8
ESP32 GND	GND rail	9
ESP32 3V3	3.3 V rail	9
ESP32 5V	5 V rail	9
Motor driver		
JP1, JP2	position 3	10
J1 pin 5 (GND)	GND rail	11
J1 pin 6 (VCC)	3.3 V rail	11
J1 pin 1 (EN1) · pin 2 (DIR1)	ESP32 13 · ESP32 14	12
J1 pin 7 (EN2) · pin 8 (DIR2)	ESP32 32 · ESP32 33	12
Left motor red · black	J5 M1+ · J5 M1–	13
Right motor red · black	J6 M2+ · J6 M2–	13
Ultrasonic sensor		
HC-SR04 GND	GND rail	15
HC-SR04 TRIG	ESP32 25	15
Resistors	a30–a35 · b35–b40 · c40–c45	16
HC-SR04 ECHO	d30	17
d35	ESP32 26	17
d45	GND rail	17
HC-SR04 VCC	5 V rail	18
Power		
Battery red + regulator IN+	J4 VM	19
Battery black + regulator IN–	J4 GND	19
Regulator OUT–	GND rail	21
Regulator OUT+	5 V rail, only with USB unplugged	29

If it doesn't work

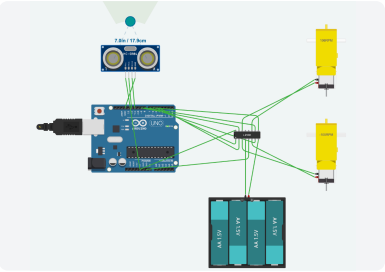
PROBLEM	TRY
No port in the Arduino IDE	Use a data cable, not a charge-only one. Install the CH340 USB driver. Try another USB port.
Upload stops at “Connecting…”	Hold the BOOT button on the ESP32 until the upload starts.
Distance always 0	Check TRIG on 25, ECHO into d30, d35 to 26, VCC on the 5 V rail, GND on the GND rail. Still 0: on this camera board pins 25 and 26 also run to the camera socket, so gently unclip the camera ribbon.
Distance jumps around	Mount the sensor level, nothing just in front of it. Test against a flat, hard surface; soft things absorb sound.
Wheels don't turn in test 2	Batteries in and switched on. Battery red on J4 VM. Terminal screws tight. J1 VCC on the 3.3 V rail. EN wires on 13 and 32.
A wheel turns the wrong way	Set `FLIP_LEFT` or `FLIP_RIGHT` to `true`, or swap that motor's two wires in its terminal.
Both wheels always move together	Move JP1 and JP2 to position 3.
ESP32 restarts when the motors start	Fresh batteries. Check every GND wire is on the GND rail and the two blue lines are joined.
Robot still hits things	Raise `STOP_DISTANCE` or lower `SPEED`. Thin, low or soft objects are hard for the sensor to see.

EXTRAS

+ Tinkercad and challenges

Try it in Tinkercad

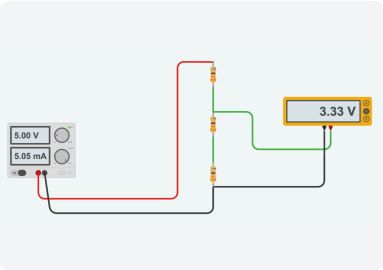
Sign in to Tinkercad (a free account), then scan a code or click a link and press **Start Simulation**. Tinkercad has no ESP32 or Pmod DHB1, so these use an Arduino Uno and an L293D motor-driver chip; the idea and the code logic are the same. To edit one, click **Copy and Tinker**.



Obstacle-avoiding robot

Click the sensor and drag the ball. Closer than 20 cm, the wheels back up and spin. The Serial Monitor shows each reading.

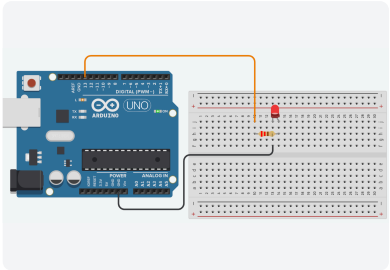
tinkercad.com/things/0d2zOu9QF5I-robosoc-w5-obstacle-avoiding-robot



Echo voltage divider

5.00 V in, 3.33 V at the tap: the circuit from steps 16 and 17. Change a resistor and watch the meter.

tinkercad.com/things/5TbZ0p51byW-robosoc-w5-echo-voltage-divider



Blink an LED

Activity 1 from the slides. The LED blinks every half second and the Serial Monitor prints on, off.

tinkercad.com/things/6x7NH8qs1pd-robosoc-w5-blink-an-led

Challenges

A. Tune it

Fastest clean run through an obstacle course. Change `SPEED`, `STOP_DISTANCE` and `TURN_TIME`.

B. Random turns

Turn left or right at random so it doesn't get stuck in corners. Replace the spin in `avoidObstacles()`:

```
if (random(2) == 0)
  drive(SPEED, -SPEED);
else
  drive(-SPEED, SPEED);
```

C. Slow down

Drive slower as things get closer. Use `s` in place of `SPEED` for forward:

```
int s = map(distance,
  20, 100, 120, 255);
s = constrain(s, 120, 255);
```